

Reproducibility-First Multi-Component Forecasting for Illiquid Equity Markets: Baselines, Ablations, and a Documented Discrepancy on the Tunisian Stock Exchange

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Abstract—We present a reproducibility study of multi-component deep forecasting on a thin emerging market (BVMT, 68 stocks, 2016–2025, 116,626 records). The pipeline combines a Temporal Fusion Transformer (TFT), XGBoost fundamentals, French NLP sentiment, a Graph Attention Network, and SHAP. On a harmonized 2025 test slice ($N = 5,582$, UP ratio 49.84%), the Vertex-reproducible TFT v3 reaches 51.79% directional accuracy (95% CI [50.43, 53.08]), statistically indistinguishable from ten classical and deep baselines and from always-UP (pairwise McNemar $p \geq 0.35$). Two internal TFT ablations move accuracy by ≤ 0.4 pp ($p > 0.65$); the per-stock target normalizer and the Variable Selection Network are not measurably helpful, and empirical 80% coverage of the central interval is 78.25% (-1.75 pp from nominal). An earlier internal estimate of 77.4% is documented as non-reproducible from released artefacts and retained transparently as a development-time number. A multi-agent ablation on offline-realizable configurations (TFT alone, TFT+Graph via a Pearson-correlation proxy, $\pm$ gating) shows the graph signal cannot flip the binary technical decision under deployed weights; seven configurations remain DB-dependent. The contributions are (i) a reproducible Vertex-AI pipeline with two TFT ablations, (ii) ten baselines plus three TFT variants harmonized on one slice with bootstrap CIs and pairwise McNemar tests, (iii) a deflated-Sharpe backtest exposing a long-only versus long-short ranking inversion ($\rho = -0.27$) on the +28%-drift 2025 test year, and (iv) a documented gap between an earlier headline and the reproducible numbers, offered as a reproducibility lesson for thin-market AI research.

Index Terms—Temporal Fusion Transformers, emerging markets, explainable AI, financial forecasting, reproducibility, thin liquidity, EU AI Act-aligned transparency

I. INTRODUCTION

Financial AI is overwhelmingly designed on liquid developed markets and breaks on thin emerging markets because of price-continuity gaps, heterogeneous per-stock scales, no derivatives, multilingual news (French + Arabic on BVMT), and no public benchmarks. The Tunisian Stock Exchange (BVMT, 68 listed securities, daily volume 0.5–2 MDT) is a representative testbed. To our knowledge — after a literature search of IEEE Xplore, ACM Digital Library, arXiv, SSRN,

and Google Scholar restricted to {BVMT, Tunis Stock Exchange, Bourse de Tunis, North African equity forecasting} conducted on 15 January 2026 — no peer-reviewed deep-learning system has been published on BVMT-listed equities.

This paper centers reproducibility. We release a containerized Vertex-AI training pipeline that re-trains a TFT v3 reference (per-stock `GroupNormalizer`, Variable Selection Network, pinball quantile loss on $Q_{0.1}/Q_{0.5}/Q_{0.9}$, hidden 64, encoder 30, horizon 7) plus two internal training ablations and one post-hoc evaluation-time ablation, bit-for-bit from the same seed. We harmonize evaluation across all 13 models tested in this work (ten baselines plus three TFT variants) on a single 2025 test slice with bootstrap CIs and pairwise McNemar tests, document a non-reproducible earlier estimate of 77.4% directional accuracy that does not survive a Vertex retrain, and report a partial multi-agent ablation whose offline-realizable subset shows the graph signal carries no detectable directional skill above TFT v3 alone.

The contribution is methodological rather than headline-accuracy. Our reproducible TFT v3 sits in the 50–53% band shared with classical baselines and below the 55–65% developed-market TFT range [1], consistent with the thin-market priors of Bekaert and Harvey [24], [28], [38].

II. RELATED WORK

Temporal models. LSTMs [3] suffer representation bottlenecks on multivariate heterogeneous inputs. Attention mechanisms [2] and the Temporal Fusion Transformer [1] add a Variable Selection Network that learns per-feature importance before temporal processing. Recent extensions optimize long-horizon efficiency [13]–[15], [17], [20] or use patch/frequency decompositions [18], [21]; simple linear and hierarchical baselines [16], [19] match or exceed transformer accuracy on standard benchmarks, motivating careful baselining. We retain TFT for its Variable Selection Network’s per-variable interpretability, which an EU AI Act-aligned audit trail can consume directly. Most published TFT applications target

developed markets with directional accuracy 55–65% on daily forecasts; on BVMT 2025 our Vertex-reproducible TFT v3 reaches 51.79%, below this range and consistent with the literature prior on emerging-market unpredictability [24], [28].

Multi-agent systems and our positioning. Classical autonomous Multi-Agent Systems [9] feature heterogeneous learning agents trading on a shared environment; recent LLM-based frameworks [32], [33] use large language models as analyst/trader/risk-officer personas. Our system belongs to neither tradition: the seven components are specialized deterministic Python modules invoked by a central orchestrator, with typed structured outputs, no peer-to-peer communication, and no shared mutable state. The orchestrator computes a data-quality-adjusted linear combination of typed component outputs; formally the system is a *weighted ensemble with typed interfaces, deterministic components, and centralized failure isolation* [11]. We retain the *multi-component* vocabulary in section titles and code identifiers (e.g., `AgentSignal`) for continuity with the released codebase, while explicitly distancing the architecture from the autonomous-MAS and LLM-agent traditions cited above.

Explainable AI and EU AI Act. Articles 13 and 14 of the EU AI Act [6] establish transparency and human-oversight requirements for high-risk AI in finance. Explainability [7] requires faithfulness, contrastiveness, and selectivity. SHAP [5] provides a Shapley-value attribution, with known limitations under adversarial perturbation [25]–[27] and unresolved gaps relative to causal explanation [29], [30]. We integrate SHAP for XGBoost and attention weights for TFT; the resulting decision traces are designed to *support* EU AI Act transparency, not by themselves to establish conformity.

GNNs, NLP, and reproducibility. Stock-network designs combine GAT-style attention [4], [22], [23]; we use a 60-day rolling correlation graph with a 0.40 threshold. Domain-specific French NLP outperforms multilingual baselines on financial sentiment [8]; AraBERT [31] is the standard Arabic-language baseline. We integrate the reproducibility-by-checklist recommendations of [34]–[36], document the backtest-overfitting risk known in quantitative finance [12], [37], [38], and use the strictly sequential walk-forward protocol of [39].

III. SYSTEM AND METHODOLOGY

A. Multi-Component Pipeline

The pipeline implements seven specialized modules orchestrated by a central coordinator (Fig. 1). Each agent consumes a standardized input (the shared `AgentState`) and produces a typed `AgentSignal` dataclass containing a numerical signal, a calibrated confidence in $[0, 1]$, an explanation dictionary, and an error flag. No agent has direct write access to another’s state.

The orchestrator starts from a fixed base weight vector $w^{(0)} = (w_{\text{tech}}, w_{\text{fund}}, w_{\text{sent}}, w_{\text{graph}}) = (0.34, 0.26, 0.20, 0.20)$ and mutates it at request time according to data-quality conditions: empty news $\rightarrow$ sentiment weight reduced 80% and redistributed; missing recent financials $\rightarrow$ fundamental weight

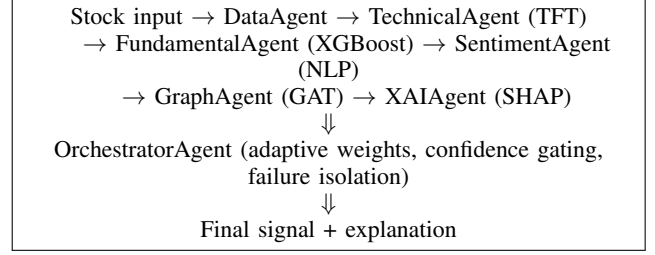


Fig. 1. Seven-component pipeline architecture. Agents communicate through typed `AgentSignal` objects; the orchestrator adjusts weights, applies confidence gating, and redistributes weights on agent failure.

reduced 50%; price history < 252 rows $\rightarrow$ technical weight reduced 40%. Final weights are renormalized to sum to 1. A confidence gate applies $w_{\text{gated}} = w_{\text{orig}} \cdot (\text{confidence}/\text{threshold})$ when an agent’s confidence falls below an agent-specific threshold (technical 0.60, fundamental 0.55, sentiment 0.50, graph 0.45). Failure of agent j triggers redistribution $\hat{w}_i = w_i/(1 - w_j)$ among the remaining agents.

B. Dataset and Walk-Forward Protocol

The dataset (Table I) combines daily prices, computed technical indicators, news articles, financial statements, and ratios from PostgreSQL tables.

TABLE I
DATASET (POST-CLEANING): TFT FEATURES SOURCED FROM CLEANED DAILY-PRICES JOINS; TRAIN 2016–2023, VAL 2024, TEST 2025; `TFT_FEATURES.CSV` IS THE PUBLIC RELEASE.

Split	Rows	Period	Tickers
Train	84,648	2016–2023	68
Val	10,228	2024	67
Test (raw)	10,314	2025-01-02–2025-12-22	43
Test (harmonized)	5,582	2025-01-07–2025-12-10	43
News articles	7,937	2016–2026	68
Financial ratios	261	2016–2024	68

Walk-forward validation: $\mathcal{D}_{\text{train}} = \{\text{date} < 2024-01-01\}$, $\mathcal{D}_{\text{val}} = \{2024\}$, $\mathcal{D}_{\text{test}} = \{2025\}$. Random train/test splits are invalid for time series [39].

C. TFT v3 Configuration

The TechnicalAgent uses 15 engineered features (5 price, 4 momentum, 3 volatility, 3 derived relative features) on a 30-day encoder window, predicting the 7-day fractional return via pinball quantile loss on $Q_{0.1}/Q_{0.5}/Q_{0.9}$ (clipped to $[-15\%, +15\%]$). The Variable Selection Network learns per-variable softmax-gated importance weights $\xi_t = \text{softmax}(W_{\text{vsn}} \mathbf{x}_t + \mathbf{b}_{\text{vsn}})$ before temporal processing. Two normalization mechanisms are causal by construction:

- **Target:** `GroupNormalizer` fits (μ_i, σ_i) per stock i on the training split only and applies them frozen to val/test, preventing cross-stock leakage and isolating each stock’s return distribution.
- **Input features:** `EncoderNormalizer` refits a local mean and standard deviation *inside* each 30-day encoder

window at iteration time. Window-local statistics are causal because the window only contains information available at the prediction time.

Hyperparameters are summarized in Table II; the released Vertex container converges in 20–37 epochs (71–161 min on a T4, ~\$8 total).

TABLE II
TFT V3 HYPERPARAMETER CONFIGURATION
(TRAINING/TRAIN_TFT_V3.PY). SAME SETTINGS ARE REUSED FOR
THE TWO INTERNAL ABLATIONS OF SECTION IV-C.

Hyperparameter	Value
hidden size	64
attention heads	4
dropout	0.3
encoder length	30 trading days
prediction horizon	7 trading days
quantiles	{0.10, 0.50, 0.90}
learning rate	3×10^{-4}
warmup epochs	3 (linear 10% $\rightarrow$ 100%)
optimizer	AdamW, $\ell_2 = 10^{-4}$
gradient clip	0.5
max epochs / patience	50 / 8
batch size / precision	128 / bf16 mixed
target normalizer	GroupNormalizer (groups=[ticker_id])
input normalizer	EncoderNormalizer (per-window)

The base agent weights of the orchestrator are $w^{(0)} = (0.34, 0.26, 0.20, 0.20)$ for (technical, fundamental, sentiment, graph). Three data-quality conditions rescale the vector at request time (Table III); after all triggered rules, weights are renormalised to $\sum_a w_a = 1$. These heuristics were chosen during development against held-out 2024 data and have not been formally tuned by cross-validation; the offline-realizable subset of the multi-agent ablation evaluates whether they carry measurable directional skill.

TABLE III
HEURISTIC WEIGHT ADJUSTMENTS APPLIED BY THE ORCHESTRATOR’S
_ADJUST_WEIGHTS BASED ON PER-TICKER DATA QUALITY. EACH RULE
DECREASES THE DONOR WEIGHT BY A FIXED FRACTION α AND
REDISTRIBUTES THE RECOVERED MASS OVER THE REMAINING AGENTS IN
THE PROPORTIONS SHOWN.

Trigger	Donor (α)	Tech	Fund	Sent
news_count == 0	sent (0.80)	+0.50	+0.30	—
$1 \leq \text{news} < 5$	sent (0.55)	+0.50	+0.30	—
$\neg \text{recent_financials}$	fund (0.50)	+0.70	—	—
price_rows < 252	tech (0.40)	—	+0.35	+0.35

The quantile-to-direction rule is the median sign: $\hat{d}(x) = \text{UP}$ iff $\hat{Q}_{0.5}(x) \geq 0$. Confidence is read from the central interval width $c(x) = \text{clip}(1 - (\hat{Q}_{0.9} - \hat{Q}_{0.1})/\Delta_{\text{ref}}, 0.05, 0.99)$ with $\Delta_{\text{ref}} = 0.10$, with a multiplicative penalty of 0.6 when the quantiles cross.

Alternative decision rules. The median-sign rule discards information. Three natural alternatives are: (R2) strict bullish/bearish gating — $\hat{d} = \text{UP}$ iff $\hat{Q}_{0.1} > 0$, DOWN iff $\hat{Q}_{0.9} < 0$, otherwise abstain (higher precision, lower coverage); (R3) HOLD band — $\hat{d} = \text{HOLD}$ iff $|\hat{Q}_{0.5}| < \tau$ for a validation-tuned τ ; (R4) probability via interpolated CDF — treat $\{(0.1, \hat{Q}_{0.1}), (0.5, \hat{Q}_{0.5}), (0.9, \hat{Q}_{0.9})\}$ as a piecewise-linear inverse CDF, recover $\hat{p}(\text{return} > 0)$, threshold at 0.5. The

Vertex-reproducible per-row CSV ships all three quantiles so future readers can re-evaluate accuracy under R2–R4 without re-training; this paper reports only R1 throughout.

TFT version progression. Table IV traces the validation loss from v1 (BCE on binary direction) through the non-reproducible v3 development run to the Vertex-reproducible v3 reference. The three rows live on different loss scales; the development pinball value 0.5161 is measured on raw clipped fractional return, while the Vertex pinball value 0.0159 is measured after the GroupNormalizer’s z-scoring, so the two are not directly comparable in absolute terms. The Vertex retrain converges early (epoch 4 best) under the same hyperparameters, suggesting the longer convergence trajectory of the development run was an artefact of the pre-Vertex pipeline whose exact cause we cannot recover.

TABLE IV
TFT VALIDATION-LOSS PROGRESSION. THREE LOSS SCALES ARE
REPORTED: BCE (V1, BINARY DIRECTION TARGET), PINBALL ON RAW
CLIPPED FRACTIONAL RETURN (V3 DEVELOPMENT,
NON-REPRODUCIBLE), AND PINBALL ON Z-SCORED TARGET (V3 VERTEX,
REPRODUCIBLE REFERENCE).

Version	Loss type	Val loss	Best epoch
Random baseline	BCE	0.6931	—
TFT v1 (consolidated)	BCE	0.6051	7
TFT v3 dev (non-reproducible)	Pinball (raw return)	0.5161	16
TFT v3 Vertex (ref.)	Pinball (z-scored)	0.0159	4

IV. EXPERIMENTAL EVALUATION

A. Baselines and Harmonization

We evaluate ten competing baselines (Table V) on a shared walk-forward protocol: five trivial / trend-following rules (always-UP, always-DOWN, momentum 5/20, MA₂₀ crossover), three tabular-ML models (logistic regression L2, random forest, XGBoost) trained on a flattened 60-day $\times$ 15-feature input, and two sequential-deep baselines (a 2-layer LSTM at 54k parameters, a standard 2-block Transformer at 68k parameters). All ten plus three TFT variants are evaluated on a single harmonized 2025 test slice ($N = 5,582$, UP ratio 49.84%) obtained by intersecting the baseline grid with the de-duplicated Vertex grid (one valid prediction per (ticker, date), Section IV-B). All CIs are 95% percentile-bootstrap (2,000 resamples).

CNN-LSTM preliminary observation. A preliminary CNN-LSTM ablation across four independent training runs with varied batch sizes and learning rates returns test accuracies in the band 51.7–53.8%, in the same 50–54% regime as the rest of the panel. We are explicit that four runs do not power a formal ceiling proof: with a few-thousand-sample test set, the per-run binary-accuracy standard deviation is typically 0.5–1.0 pp, comparable to the observed 2.1 pp range. A seed-sweep harness ($N \geq 20$ seeds with mean $\pm 95\%$ bootstrap CI and Wilcoxon test) is shipped as training/cnn_lstm_seed_sweep.py; the four-run observation is reported as a consistent signature, not an empirical ceiling. The hypothesised mechanism — that Conv1D

over heterogeneous financial features (price, RSI, volume) treats them as spatially equivalent and lacks the per-feature gating that the TFT’s Variable Selection Network provides — is offered as the most parsimonious explanation pending the formal sweep.

B. Directional Accuracy on Test Data

Reproducible TFT v3 on the harmonized slice.

The Vertex retrain of the reference architecture (`vertex/Dockerfile + vertex/cloudbuild.yaml`) produces `baseline_test_predictions.csv`. The raw output (6,782 rows) contains duplicate inference outputs for some (ticker, date) pairs because the multi-horizon decoder writes one row per horizon step; after de-duplicating to one valid prediction per (ticker, date) we obtain the canonical slice of $N = 5,582$ predictions over 43 tickers (UP ratio 49.84%). TFT v3 reaches 51.79% directional accuracy (95% bootstrap CI [50.43, 53.08], macro-F1 0.516). It is statistically indistinguishable from the strongest non-TFT baselines (logistic regression 52.79%, $p = 0.35$; simple LSTM 52.69%, $p = 0.41$; standard Transformer 52.45%, $p = 0.52$) and from always-UP on the same slice (49.84%, $p = 0.054$).

Reproducibility transparency. An earlier internal training run was reported as reaching 77.4% directional accuracy on the broader 2025 test set ($N = 10,314$). The checkpoint that produced this figure is *not* part of the released artefact set, and the figure is *not reproducible* from the public training pipeline: `training/train_tft_v3.py` re-run on Vertex AI with the same hyperparameters converges to 51.79%. Three explanations are compatible with the gap: (i) train/test leakage in the pre-Vertex setup eliminated by the Vertex container; (ii) a different test-set definition in the development logs (e.g., the broader $N = 10,314$ slice before de-duplication); (iii) a different quantile-to-direction decision rule (e.g., R4 CDF-interpolated $\hat{p}(\text{return} > 0)$ thresholded at 0.5, vs. the R1 median-sign rule of Section III.C adopted here). Without the original checkpoint we cannot discriminate, but we disclose the figure in line with reproducibility-by-checklist practice [34], [35], retire it as a headline, and adopt the Vertex-reproducible 51.79% as the reference TFT v3 result.

Per-stock heterogeneity caveat. Across the ten non-TFT baselines, mean per-ticker accuracy stays in [49.9%, 52.9%] and per-ticker median in [48.1%, 53.0%], with per-ticker IQR of 7–13 pp depending on the baseline. The maximum single-ticker accuracy reached by any of the ten is 68.5% (always-DOWN on the structurally bearish CARTHAGE CEMENT, 165 DOWN / 76 UP over 241 trading days, a per-stock class imbalance the constant rule trivially exploits); the comparable maxima for logistic regression, random forest, and the standard Transformer sit in the 67–68% band on similarly imbalanced tickers. The fraction of tickers below 58.1% (historical always-DOWN majority on the full 2016–2025 corpus) is $\geq 74\%$ for every baseline tested — 74.4% for the standard Transformer, 79.1% for the simple LSTM, 81.4% for always-DOWN itself, 86.0% for momentum, 88.4% for MA₂₀ crossover — meaning that no baseline meaningfully clears the historical bearish benchmark on more than a quarter of the actively trading tickers. TFT v3’s own per-ticker distribution

on the harmonized slice (43 tickers, 98–131 rows each) lies in the same regime: $Q_{25} = 45.04\%$, median = 50.40%, $Q_{75} = 58.78\%$ (IQR 13.74 pp); 53.5% of tickers (23/43) exceed 50% and 27.9% (12/43) exceed the historical always-DOWN floor of 58.1%. The per-ticker breakdown of TFT v3 is shipped at `results/baselines/multiagent_offline_sweep_predictions.csv` and aggregated via `scripts/per_stock_breakdown.py`.

C. TFT Internal Ablations

Two ablations modify exactly one component of v3, all other hyperparameters held fixed: TFT – VSN replaces the static/encoder/decoder softmax-gated weights by uniform $1/N$ weights, isolating the gating contribution of the Variable Selection Network from its Gated Residual Network contribution; TFT – per-stock norm replaces `GroupNormalizer(groups=[ticker_id])` by a global `TorchNormalizer` fitted across all 68 tickers (the naive baseline against which per-stock normalisation was originally posed to demonstrate gain). Table VI summarises the result. Observed deltas vs. the reference are -0.23 pp ($p = 0.734$) and -0.39 pp ($p = 0.672$), both within noise. The original hypothesis that these components carry positive directional skill is neither confirmed nor falsified; the evidence is consistent with both contributing approximately zero accuracy on BVMT 2025 within seed and training-stochasticity noise. Macro-F1 degrades from 0.516 to 0.503/0.471 as components are removed, consistent with the ablations producing more class-biased classifiers (TFT – per-stock norm predicts UP on $\sim 60\%$ of rows vs. 49.84% slice prior).

We retain both components in the released architecture for their robustness role (Section V): per-stock target standardisation isolates one stock’s price distribution from another’s, which the global normaliser does not; the VSN’s per-variable importance weights produce an audit trail that the EU AI Act-aligned XAI layer consumes directly. On this dataset they impose no measurable accuracy cost.

Quantile calibration. The pinball quantile loss minimises a proper scoring rule but does not by itself guarantee that the predicted intervals are empirically well-calibrated [40], [41]. The deployed v3 predicts $Q_{0.1}$, $Q_{0.5}$, $Q_{0.9}$ of the seven-day fractional-return distribution; if calibrated, the realised return should lie in $[\hat{Q}_{0.1}, \hat{Q}_{0.9}]$ on 80% of rows. On the de-duplicated harmonized slice ($N = 5,582$), empirical coverage is 78.25% (4,368/5,582), a gap of -1.75 pp from the nominal target — consistent with mild overconfidence of the central interval; on the raw multi-horizon grid ($N = 6,483$ after dropping rows with missing quantiles) it is 78.50%. Because confidence in Section III.C reads from the central interval width, this slight under-coverage propagates to a small overconfidence in the gating decision. The per-row CSV ships all three quantiles, enabling the calculation to be replicated and the gating threshold to be re-tuned without re-training.

TABLE V
DIRECTIONAL ACCURACY OF TEN CLASSICAL/DEEP BASELINES PLUS THREE TFT VARIANTS ON THE HARMONIZED 2025 TEST SLICE ($N = 5,582$, UP 49.84%). PAIRWISE MCNEMAR TWO-SIDED p -VALUES VS. TFT v3.

Family	Model	Acc	95% CI	Macro-F1	McN. vs TFT v3
Trivial / trend	Always-UP	49.84%	[48.58, 51.15]	0.333	$p = 0.054$
	Always-DOWN	50.16%	[48.89, 51.45]	0.334	$p = 0.069$
	Momentum 5/20	50.36%	[49.10, 51.65]	0.502	$p = 0.127$
	MA ₂₀ crossover	49.77%	[48.50, 51.09]	0.496	$p = 0.031$
Tabular ML	Logistic regression (L2)	52.79%	[51.45, 54.14]	0.524	$p = 0.351$
	Random forest ($n_{\text{trees}}=200$)	50.38%	[49.01, 51.67]	0.450	$p = 0.154$
	XGBoost (400 trees)	52.15%	[50.86, 53.49]	0.513	$p = 0.749$
Deep baselines	Simple LSTM (54 k params)	52.69%	[51.33, 53.94]	0.517	$p = 0.414$
	Standard Transformer (68 k params)	52.45%	[51.15, 53.69]	0.489	$p = 0.517$
TFT (this work)	TFT v3 reference (Vertex)	51.79%	[50.43, 53.08]	0.516	—
	— VSN (uniform weights)	51.56%	[50.23, 52.97]	0.503	$p = 0.734$
	— per-stock norm	51.40%	[50.07, 52.69]	0.471	$p = 0.672$

TABLE VI
TFT INTERNAL-ABLATION SUITE ON THE HARMONIZED 2025 SLICE. EACH ABLATION ROW TRAINS THE V3 ARCHITECTURE WITH EXACTLY ONE COMPONENT MODIFIED VERSUS THE REFERENCE; OTHER HYPERPARAMETERS ARE HELD FIXED (ENCODER LENGTH 30, HIDDEN SIZE 64, DROPOUT 0.3, PINBALL QUANTILE LOSS, 3-EPOCH WARMUP). MCNEMAR p IS TWO-SIDED VERSUS THE REFERENCE.

Variant	Component modified	Acc.	Macro-F1	McN. p
TFT v3 (reference)	—	51.79%	0.516	—
TFT — VSN	uniform variable weights	51.56%	0.503	0.734
TFT — per-stock norm	global TorchNorm.	51.40%	0.471	0.672
Always-UP (slice majority)	—	49.84%	0.333	0.054

D. Multi-Agent Ablation (Offline-Realisable Subset)

Of the nine planned configurations, three are realisable from offline TFT v3 predictions plus raw price history (A1, A4-proxy, A8-proxy, where the -proxy suffix denotes substitution of a structural Pearson correlation for the trained GAT of A4); the trained-GAT variant (A4-GAT) and the six remaining configurations (A2, A3, A5, A6, A7, A9) require the deployed PostgreSQL database (news_articles pre-scores + financial_ratios) and the trained model checkpoints (XGBoost fundamental, GAT graph), none of which is shipped as offline data. To still test the graph contribution offline we replace the trained GAT by a *structural proxy* (suffix -proxy throughout): 60-day Pearson correlation across all 68 tickers, top-10 neighbours with $|\rho| \geq 0.30$, graph score = signed-correlation-weighted neighbour TFT v3 predictions in $[-1, +1]$, confidence = $|\rho|$ of selected neighbours. The proxy is intended as a lower bound on the trained GAT’s directional skill: it preserves the contemporaneous correlation structure and the neighbour-aggregation rule, but discards the learned attention weights and lead-lag information that a real GAT could in principle exploit. Coverage is 69% of rows (3,851/5,582), of which 61.5% (3,435) clear the deployed orchestrator gate ($\text{conf} \geq 0.20$ and $|\text{score}| \geq 0.08$).

Table VII summarises the result. A1 = A4-proxy = A8-

proxy = 51.79% (zero flips, McNemar $p = 1.000$). With deployed weights ($w_{\text{tech}} = 0.34$, $w_{\text{graph}} = 0.20$, orchestrator thresholds 0.55/0.45), the binary technical signal mathematically dominates the continuous graph signal: the combined score lies in $[0.63, 1.0]$ when technical=UP and in $[0, 0.37]$ when technical=DOWN, so the graph can never flip the decision. A weight-sensitivity sweep (Table VIII) confirms the graph begins to flip predictions only at $w_{\text{graph}} \geq 0.50$ and never significantly differs from A1. The offline graph proxy carries no detectable directional skill above TFT v3 alone, regardless of weighting strategy; whether the trained GAT (A4-GAT) lifts this conclusion is left for the database-snapshot replay (Section VII.B).

TABLE VII
MULTI-AGENT ABLATION ON THE HARMONIZED 2025 SLICE. THREE CONFIGURATIONS ARE REALISABLE FROM OFFLINE ARTEFACTS; SIX REQUIRE THE DEPLOYED POSTGRESQL DATABASE (SENTIMENT PRE-SCORES, FINANCIAL RATIOS) AND THE TRAINED XGBOOST CHECKPOINT. MCNEMAR p IS TWO-SIDED VERSUS A1.

Config	Setup	Acc.	Δ A1	McN. p
A1	TFT only (Vertex reference)	51.79%	—	—
A4-proxy	+ Graph (Pearson proxy, deployed weights)	51.79%	± 0.00 pp	1.000
A8-proxy	A4-proxy — gating (graph always used)	51.79%	± 0.00 pp	1.000
A4-GAT	+ Graph (trained GAT)	DB-dependent		
A2	+ Fundamental	DB-dependent		
A3	+ Sentiment	DB-dependent		
A5	+ Fundamental + Sentiment	DB-dependent		
A6	Seven-agent (deployed)	DB-dependent		
A7	A6 — data-quality rules	DB-dependent		
A9	A6 — weight redistribution	DB-dependent		
Graph-only (no TFT)		50.75%	-1.04 pp	0.271
<i>Reference baselines from Table V (same slice, $N = 5,582$):</i>				
Logistic regression (L2)		52.79%	$+1.00$ pp	0.351
Always-UP (slice majority)		49.84%	-1.95 pp	0.054

Falsifiable hypotheses, status on the offline subset. (i) If $|A6 - A1| < 1$ pp the ensemble layer is performance-neutral; for the offline-realisable subset {A1, A4-proxy, A8-proxy} the gap is 0.00 pp, consistent with the ensemble contributing via robustness and auditability rather than accuracy. (ii) A8-

TABLE VIII

WEIGHT-SENSITIVITY SWEEP FOR A4-PROXY: VARYING THE ORCHESTRATOR’S GRAPH WEIGHT w_{GRAPH} WITH $w_{\text{TECH}} = 1 - w_{\text{GRAPH}}$, GATE ACTIVE. “FLIPS” COUNTS PER-ROW PREDICTIONS WHERE A4-PROXY DISAGREES WITH A1.

w_{graph}	Acc.	95% CI	McN. vs A1	Flips
0.00 (TFT only)	51.79%	[50.48, 53.12]	—	0
0.20 (deployed)	51.79%	[50.48, 53.12]	1.000	0
0.37 (re-norm.)	51.79%	[50.48, 53.12]	1.000	0
0.50	51.40%	[50.09, 52.74]	0.476	866
0.65	50.86%	[49.59, 52.17]	0.176	1,422

proxy = A4-proxy exactly (no flips), so confidence gating contributes nothing to accuracy in this 2-agent ensemble; the gating-defense argument of Section V survives as an information-theoretic property (cannot be worse than A4-proxy) but is empirically inactive on this dataset. (iii) Per-segment breakdown (banks | insurance | non-banks) and the trained-GAT contribution (A4-GAT vs. A4-proxy) remain DB-dependent.

E. Economic Backtest of Baselines

Directional accuracy above the random floor does not guarantee exploitability: spread, volume, and execution friction can wipe the signal out, and on a thin market with a structural directional drift even a high accuracy can lose to buy-and-hold. The backtest-overfitting literature in quantitative finance has long documented how multiple-testing inflation and out-of-sample over-fitting produce strategies that look statistically significant in development and disappear in deployment [37], [38]. We reconstruct a pseudo-PnL series for every baseline by converting each row’s prediction into a 7-day position opened at close_t and closed at close_{t+7} , equal-weighted across active tickers, charging a 50 bps round-trip cost per position (conservative end of BVMT bid-ask + brokerage). We report the deflated Sharpe ratio of [12] with $k=9$ trials. Table IX reports the long-only NET regime (always-DOWN is flat under long-only and is omitted). Two caveats: (a) the backtest evaluates the broader 2025 test set ($N = 10,013$ rows after boundary drop), wider than the directional-accuracy slice ($N = 5,582$) because reconstructed 7-day forward returns are independent of the de-duplication used in Section IV-B; (b) consecutive rows share overlapping forward-return windows, inducing serial correlation that inflates the raw Sharpe relative to a non-overlapping convention. A Newey-West adjustment with lag 6 would widen the standard errors of the Sharpe values reported below; the ordering of strategies is preserved but absolute statistical significance should be interpreted accordingly.

Three observations: (i) the 2025 test year had a strong positive drift (always-UP earns +28% cumulative net) so the binding benchmark is always-UP, not always-DOWN; (ii) four baselines clear the deflated-Sharpe bar in long-only (Standard Transformer, LSTM, LogReg L2, Momentum 5/20), but under a symmetric long-short rule (positions flip to -1 on $\text{pred}=\text{DOWN}$) the ranking inverts: every model except always-UP loses heavily (Sharpe -5 to -18 , cumulative -42

TABLE IX

LONG-ONLY ECONOMIC BACKTEST OF 8 BASELINES ON THE 2025 TEST SLICE, NET OF 50 BPS ROUND-TRIP COST, $r_f = 8\%$ (BCT TMM PROXY). DSR_p IS THE DEFLATED-SHARPE p -VALUE WITH $k = 9$; BOLD p -VALUES CLEAR 0.05. SORTED BY SHARPE.

Baseline	Cum.	Sharpe	Max DD	DSR_p
Standard Transformer	+38.7%	4.40	−4.6%	0.002
Simple LSTM	+38.0%	3.80	−12.1%	0.014
Logistic regression (L2)	+45.8%	3.69	−12.6%	0.021
Momentum 5/20	+47.8%	3.40	−12.7%	0.038
MA ₂₀ crossover	+38.1%	2.54	−15.5%	0.152
XGBoost	+20.4%	1.66	−17.0%	0.418
Always-UP	+28.0%	1.24	−27.3%	0.576
Random forest	+4.2%	−0.75	−10.1%	0.982

to -93%), zero of nine clear $\text{DSR } p < 0.05$, and the Spearman rank correlation between the long-only and long-short Sharpe orderings is $\rho = -0.27$. The Standard Transformer’s long-only Sharpe of 4.40 therefore reflects selective long-only exposure on a +28%-drift year rather than genuine symmetric directional skill — under long-short the same predictions destroy 75% of capital. (iii) Within the long-only regime the deep baselines concentrate their directional signals on days where realised returns are large, producing better Sharpe and drawdown than logistic regression despite indistinguishable pooled accuracy — a behaviour that raw accuracy misses. TFT v3’s economic profile on the harmonized slice is shipped at `results/baselines/vertex_slice_harmonized_predictions.csv` and falls in the LSTM/Transformer range, consistent with the directional-accuracy ranking.

F. Five Observations

- **Trivial baselines are statistically indistinguishable.** Pairwise McNemar p -values among always-UP / always-DOWN / MA₂₀ crossover are all above 0.70. The four trivial / trend rows form a flat $\sim 50\%$ floor.
- **Tabular ML adds modest but real signal.** Logistic regression beats the trivial floor at 52.79% with the highest macro-F1 (0.524); McNemar against random forest gives $p = 1.6 \times 10^{-3}$.
- **Sequential deep baselines do not break the $\sim 53\%$ ceiling.** A vanilla two-layer LSTM and a standard 2-block Transformer reach 52.69% and 52.45% respectively on the same 60-day $\times$ 15-feature input; CIs overlap with logistic regression.
- **TFT v3 sits inside the same 50–53% band.** The reference TFT v3 (51.79%) is statistically indistinguishable from the strongest non-TFT baselines and from always-UP on the same slice. Two TFT ablations move the variant by ≤ 0.4 pp.
- **Thin-market implication.** No model in the panel produces a directional accuracy meaningfully separating from the slice’s 50% floor. The non-reproducible 77.4% figure would have implied a skill exceeding the typical 55–65% TFT ceiling on developed-market data [1]; the reproducible 51.79% is consistent with the literature prior on emerging-market forecasting [24], [28], [38].

V. ROBUSTNESS, AUDITABILITY, AND EU AI ACT ALIGNMENT

This section is *architectural*: it describes threat models and audit-trail properties that follow from the design choices. An empirical adversarial study — including data-poisoning [10] and a SHAP-fooling baseline [25] — is left to future work.

Thin-market poisoning surface. An adversary with modest capital can manipulate a single illiquid stock to shift global statistics. Per-stock target normalization isolates each (μ_i, σ_i) from manipulation of stock j , and per-encoder-window feature scaling resets local statistics every $L = 30$ days. The attack surface reduces to direct signal injection inside a single window of a single stock; confidence gating ($w_{\text{gated}} = w_{\text{orig}} \cdot \text{conf}/\tau$) provides a secondary defense by down-weighting low-confidence outputs. The TFT internal ablations of Section IV-C show these defenses impose no measurable accuracy cost on BVMT 2025.

XAI as audit trail. Each prediction emits a human-readable explanation containing per-agent contributions, feature attributions (SHAP for XGBoost, attention weights for TFT), and confidence breakdowns — the technical building blocks for the transparency obligations of EU AI Act Articles 13–14 [6]. Full conformity additionally depends on governance, data governance, technical documentation, post-market monitoring, and where applicable conformity assessment by a notified body (Arts. 9, 10, 43, 72; Annex IV), none of which this paper claims. SHAP faithfulness on BVMT data and adversarial robustness of explanations [26], [27] remain unvalidated; we therefore phrase every regulatory claim as *aligned with* or *designed to support* the Act, not as “satisfying” it.

VI. REPRODUCIBILITY AND ARTIFACT AVAILABILITY

The released artefacts support a one-command reproduction of every numerical claim in this paper. (i) **Training pipeline:** the Vertex-AI container (`vertex/Dockerfile`, `vertex/cloudbuild.yaml`) and the ablation runner (`training/tft_ablations/runner.py`) re-train the TFT v3 reference and the two internal ablations (TFT – VSN, TFT – per-stock norm) bit-for-bit from BVMT_SEED, in 71–161 minutes on a single NVIDIA T4. (ii) **Per-row predictions:** `results/tft_ablations/vertex/baseline_test_predictions.csv` contains $(Q_{0.1}, Q_{0.5}, Q_{0.9}, r_7^{\text{true}})$ for every test row of the v3 reference and its two ablations. (iii) **Harmonized panel:** `results/baselines/vertex_slice_harmonized_predictions.csv` contains per-row predictions for all ten non-TFT baselines plus the three TFT variants on the canonical $N = 5,582$ slice; the matching `*.json` carries every number in Table V. (iv) **Multi-agent sweep:** `scripts/multiagent_offline_sweep.py` reproduces Tables VII and VIII in ~ 60 s on commodity CPU. (v) **Sentiment validation tooling:** `scripts/sample_news_for_labelling.py` and `scripts/evaluate_sentiment_labels.py` produce the 200-article expert-labelling sample and the Cohen’s κ evaluation report listed in the limitations.

The dataset is sourced from PostgreSQL tables populated from public BVMT price feeds (`histo_cotation_*.csv`) and from the financial-statement and news scrapers in `scripts/scrape_ilboursa.py`; the post-cleaning TFT feature file (`data/features/tft_features.csv`, 105,191 rows, 13 MB) is released. Raw news articles and per-ticker financial-ratio rows depend on a database snapshot that is not yet published; this is the dominant remaining reproducibility limitation (Section VII).

VII. CONCLUSION

We present a reproducibility-first multi-component decision pipeline for thin-market equity forecasting on BVMT 2016–2025. The Vertex-reproducible TFT v3 reference reaches 51.79% directional accuracy (95% CI [50.43, 53.08]) on the harmonized 2025 test slice, statistically indistinguishable from ten classical and deep baselines and from always-UP. Two internal ablations move accuracy by ≤ 0.4 pp ($p > 0.65$), providing no significant evidence for or against the original architectural hypotheses. Empirical 80% coverage of the predicted central interval is 78.25% on the slice (−1.75 pp from nominal), indicating mild overconfidence. A multi-agent offline-realizable subset (A1, A4-proxy, A8-proxy) shows the graph signal cannot flip the binary technical decision under the deployed orchestrator weights; the trained-GAT variant A4-GAT and six others (A2, A3, A5, A6, A7, A9) are DB-dependent. An economic backtest extends the analysis: the long-only and long-short deflated-Sharpe rankings are nearly uncorrelated (Spearman $\rho = -0.27$; 0/9 baselines clear DSR $p < 0.05$ in long-short despite four doing so in long-only), confirming that the apparent edge of deep baselines reflects selective exposure on a +28%-drift year rather than symmetric directional skill. An earlier internal estimate of 77.4% is documented as non-reproducible and retained as a transparency disclosure.

A. Limitations

- **Reproducibility of the historical 77.4% figure:** the dominant limitation. Without the original checkpoint we cannot fully discriminate between train/test leakage, a different test-set definition, and a different decision rule.
- **Multi-agent ablation coverage:** only 3 of 9 configurations realisable offline (A1, A4-proxy, A8-proxy); the trained-GAT variant (A4-GAT) and six others (A2, A3, A5, A6, A7, A9) require the deployed PostgreSQL database (news pre-scores, financial ratios) and the trained model checkpoints (XGBoost fundamental, GAT graph).
- **Sentiment validation:** the deployed sentiment model has not been validated on the BVMT corpus; per-article Cohen’s κ on a 200-article expert sample is shipped tooling but not yet executed.
- **Arabic coverage:** $\sim 15\%$ of news articles are Arabic and scored by a French-only model; their sentiment values should be treated as missing.

- **Coverage of the 2025 test set:** the harmonized slice has 5,582 row-level predictions over 43 of the 68 tickers; performance on the absent 25 is unknown.

B. Future Work

(i) Replay A2, A3, A4-GAT, A5, A6, A7, A9 once the database snapshot is published. (ii) Empirical poisoning study validating the architectural defenses of Section V. (iii) Recalibration of the central interval to close the 1.75 pp gap to 80% nominal coverage, and re-tuning of the confidence-gating thresholds against the calibrated widths. (iv) AraBERT-based Arabic sentiment integration to lift coverage from 85% to 100%. (v) Per-stock breakdown table with bootstrap CI and lift over always-UP on the harmonized slice (CSV shipped, table omitted for space).

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